Maths

Foundations of Image Science

1. Linear vector spaces
2. Types of operators
    1. Functions and functionals
    2. Integral transforms
    3. Matrix operators
    4. Continuous to discrete mappings
    5. Differential operators
3. Hilbert space operators
    1. Range and domain
    2. Linearity, compactness, and continuity
    3. Compactness
    4. Inverse operators
    5. Adjoint operators
    6. Projection operators
    7. Outer products
4. Eigenanalysis
    1. Eigenvectors and eigenvalue spectra
    2. Similarity transformations
    3. Eigenanalysis in finite-dimensional spaces
    4. Eigenanalysis of Hermitian operators
    5. Diagonalization of a Hermitian operator
    6. Simultaneous diagonalization of Hermitian matrices
5. Singular Value Decomposition
    1. Definition and properties
    2. subspaces
    3. SVD representations of vectors and operators
6. Moore-Penrose pseudoinverse

Probability

1. Probabilistic graphical models can only do 2 things:
    1. Inference: calculating a probability of some data modelled as RV using the model $\mathcal{M}$:
        1. evaluation $P(obs|state)$,
        2. decoding $P(state|obs)$
    2. Learning model parameters: refers to finding the graphical model $\mathcal{M}$, given some data $\mathcal{D}$.
        1. MAP (maximum a posteriori)
        2. Maximum likelihood
        3. expectation maximization
        4. forward-backward algorithm
2. Conditional probability can be computed using:
    1. Bayes rule
    2. Enumeration

References:

1. Harrison H. Barrett, Kyle J. Myers. Foundations of Image Science.
2. Neural Message Passing